CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

LAB MANUAL

Production & Operations Management

(ME-403.01)

ME – 403.01 PRODUCTION & OPERATION MANAGEMENT	
CO1	Identify activities of production, operation and different production systems.
CO2	Analyze and point out different sales forecasting techniques.
CO3	Prepare the product design procedure of an existing product and judge it on the basis of other products.
CO4	Observe, measure and rearrange production planning and control activities.
CO5	Identify meaning of quality by inferring different parameters to prioritize product value to adopt new changes in a changing circumstance.
CO6	Anticipate use of modern production management tools.

List of Experiment (ME403.01 - POM)		
Sr. No.	**Title**	**Course Outcomes**
1	Product Design & Development	CO3
2	Problems on Sales Forecasting Techniques	CO2
3	Exercise on Operation and Route-Sheets	CO1,CO4
4	Problems on Sequencing Technique, Gantt chart	CO1, CO4
5	Problems on X-R Charts & O C Curve	CO4, CO5
6	Verification of Proof of Sampling	CO5
7	Operation Characteristic Curve [0.C.Curve]	CO5
8	Production and Operations Management Case-Studies	CO1, CO3
9	Case-Study on JIT IN India	CO6
10	ISO9000, 14000 CASE-STUDY	CO5

List of Assignment (ME403.01 - POM)		
Sr. No.	**Title**	**Course Outcomes**
1	Assignment 1- Production systems and Sales forecasting	CO1, CO2
2	Assignment 2- Problems on Inspection and Quality Control	CO4, CO5
3	Assignment 3- Exercise on Operation and Route-Sheets	CO1,CO4
4	Assignment 4-Problems on Sequencing Technique	CO4

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Production & Operations Management(ME-403.01) **Date:**

Experiment No. 1
PRODUCT DESIGN &DEVELOPMENT

Aim: To study product design and development

Objective:
- To understand the concept of product design
- To prepare report on design and development of any product

Theory:

Defination of Product Design:
“Conversion of ideas into reatity to fulfil human needs”
Thus it is nothing but a conversion of conception into reality through following stages.

Conception → Acceptance → Execution
↓
Translation
↓
Pre-production
↓
Freezing of the Design

Keywords:

Product: Tangible output of a production system.

Design: Translation of requirement into a form.

Research: The deliberate and planned effort to discover new ideas, techniques, systems application etc.

Development: Improvement of the existing techniques or systems.

Innovation: It is the generation of new ideas.

Prototype: It is a model of a product before going to large production of any major product

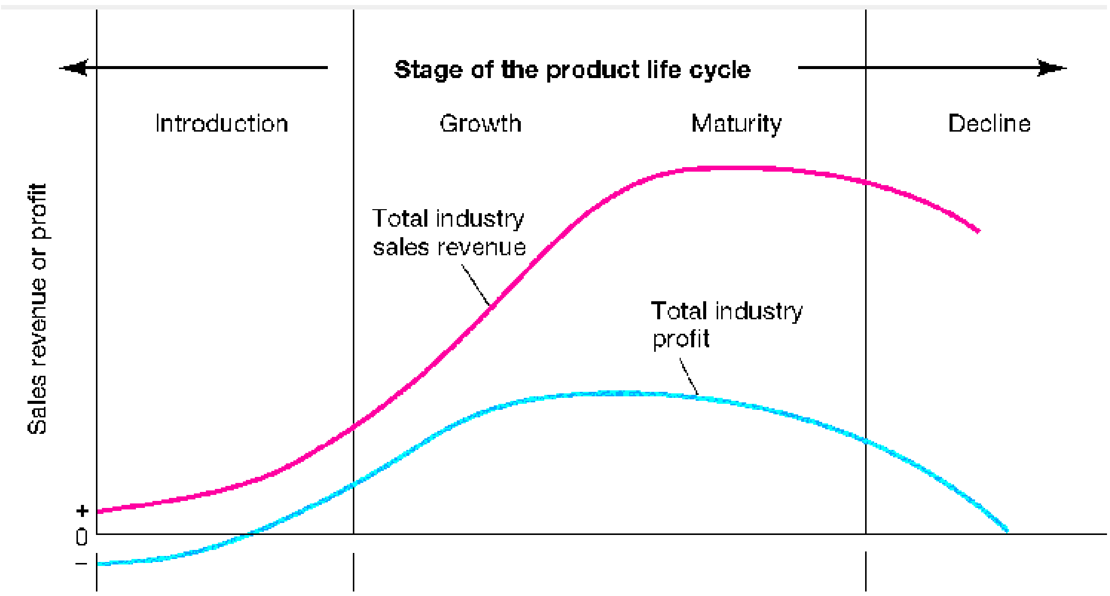
Rapid prototyping: Recent techniques which enable to produce a solid object with the required shape, size, and accuracy directly from it is computer model.

Module: It is a component made of several parts that are frequently used interchangeably between a no of product.

Robust Design: It is one that performs as intended even under non ideal conditions such as manufacturing process variation or a range of operating situations.

Patent: A temporary monopoly granted by government to an inventor to exclude other from using an invention.

Concurrent Design: A new approach to design that involves the simultaneous design of products and processes by design teams.



Stages of the product life cycle

Review Questions:

1. Define product development.
2. Explain product development procedure.
3. Explain factors affecting product design in detail.
4. Explain Standardization in detail stating its advantages and disadvantages.
5. Explain Simplification concept in detail.
6. Give an example of a product or service you have encountered that was poorly designed. Read about more bad designs at the bad design website http://www.baddesigns.com. Make a list of the factors that makes a design unworkable.

Reference Books:

1. PPC and Industrial Management by K C Jain and L N Agrawal
2. Industrial Engineering and Management by O P Khanna
3. Statistical Quality Control by Mahajan
4. PERT and CPM by R C Gupta
5. Modern Production Management by Buffia
6. Production System, Planning, analysis and Control by J L Riggs
7. Basic Marketing by Philip, Kotler
8. Elements of PPC by Samual Eilon
9. Operation Management by Roberta S. Russell

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Production & Operation Management (ME-403.01) Date:

Experiment No. 2
PROBLEMS ON SALES FORECASTING TECHNIQUES

Aim: Exercise based on Sales Forecasting.

- Problem on regression analysis
- Problems on moving average method
- Problems on exponential smoothing method
- Delphi method

Objective:

- To understand the importance of Sales Forecasting analysis in industry.
- To understand how to predict sales or demands for a product in the industry?
- To study different sales forecasting techniques.

Theory:

The growing competition, frequent changes in customer's demand and the trend towards automation demand that decisions in business should not be based purely on guesses rather on a careful analysis of data concerning the future course of events. More time and attention should be given to the future than to the past, and the question 'what is likely to happen?' should take precedence over 'what has happened?' though no attempt to answer the first can be made without the facts and figures being available to answer the second. When estimates of future conditions are made on a systematic basis, the process is called forecasting and the figure or statement thus obtained is defined as forecast.

In a world where future is not known with certainty, virtually every business and economic decision rests upon a forecast of future conditions. Forecasting aims at reducing the area of uncertainty that surrounds management decision-making with respect to costs, profit, sales, production, pricing, capital investment, and so forth. If the future were known with certainty, forecasting would be unnecessary. But uncertainty does exist, future outcomes are rarely assured and, therefore, organized system of forecasting is necessary. The following are the main functions of forecasting:

- The creation of plans of action.
- The general use of forecasting is to be found in monitoring the continuing progress of plans based on forecasts.
- The forecast provides a warning system of the critical factors to be monitored regularly because they might drastically affect the performance of the plan.

It is important to note that the objective of business forecasting is not to determine a curve or series of figures that will tell exactly what will happen, say, a year in advance, but it is to make analysis based on definite statistical data, which will enable an executive to take advantage of future conditions to a greater extent than he could do without them. In forecasting one should note that it is impossible to forecast the future precisely and there always must be some range of error allowed for in the forecast.

Dependent versus Independent Demand

Demand of an item is termed as independent when it remains unaffected by the demand for any other item. On the other hand, when the demand of one item is linked to the demand for another item, demand is termed as dependent. It is important to mention that only independent demand needs forecasting. Dependent demand can be derived from the demand of independent item to which it is linked.

Business Time Series

The first step in making a forecast consists of gathering information from the past. One should collect statistical data recorded at successive intervals of time. Such a data is usually referred to as time series. Analysts plot demand data on a time scale, study the plot and look for consistent shapes and patterns. A time series of demand may have constant, trend, or seasonal pattern (Figure 1) or some combination of these patterns. The forecaster tries to understand the reasons for such changes, such as,

- Changes that have occurred as a result of general tendency of the data to increase or decrease, known as secular movements.
- Changes that have taken place during a period of 12 months as a result in changes in climate, weather conditions, festivals etc. are called as seasonal changes.
- Changes that have taken place as a booms and depressions are called as cyclical variations.
- Changes that have taken place as a result of such forces that could not be predicted (like flood, earthquake etc.) are called as irregular or erratic variations.

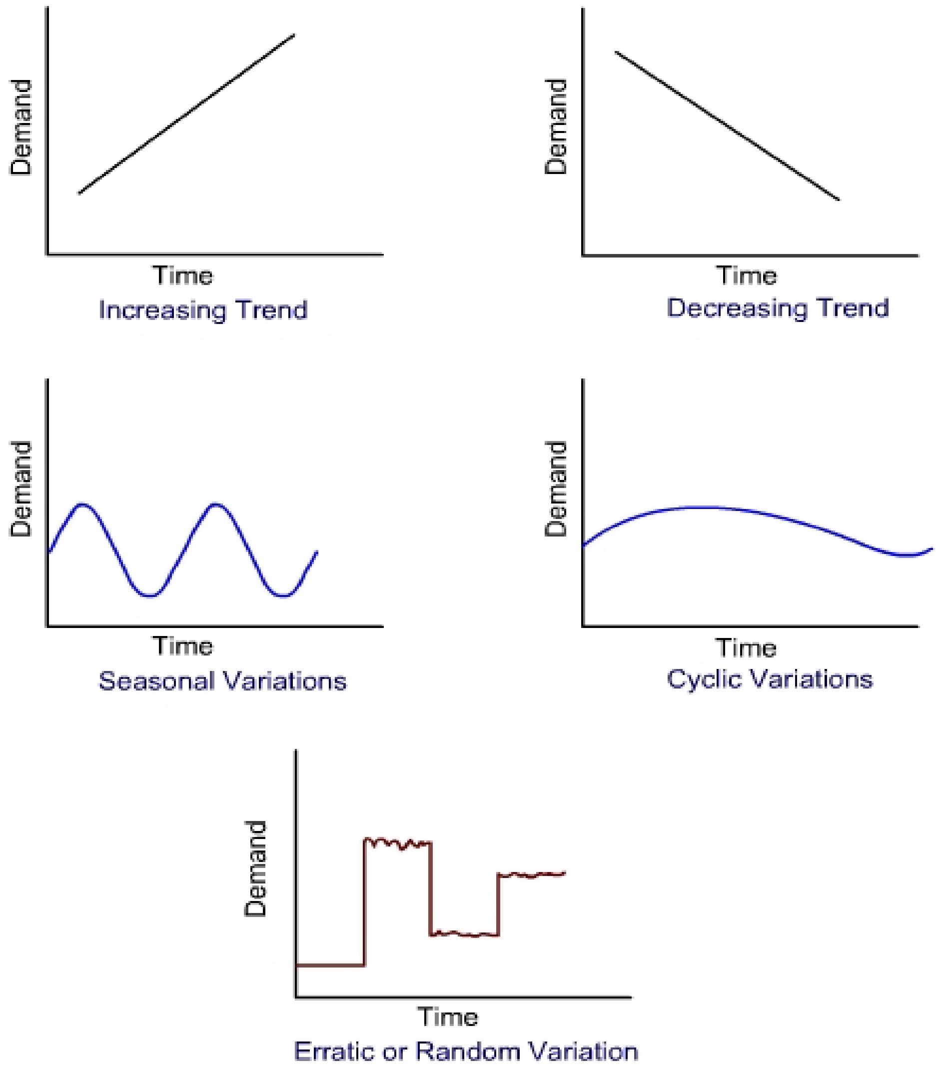


Figure: 1

Review Questions:

1. What is the difference between quantitative forecast methods and qualitative Forecast methods**?**
2. Describe difference between short range and long range Forecast
3. Describe Delphi method of forecast.
4. The following series relates to the annual sales in thousands of a product during the period 1988 – 2003. Find the trend of sales using (i) 3 Yearly moving average (ii) 5 Yearly moving average (iii) 7 Yearly moving average.

Year	Sales (* 000)	Year	Sales (* 000)
1988	16	1996	25
1989	18	1997	28
1990	15	1998	26
1991	17	1999	22
1992	20	2000	28
1993	22	2001	24
1994	25	2002	25
1995	24	2003	30

5. Calculate the forecast for a company for seventh year by exponential smoothing method from following data. Take $\alpha = 0.25$ and initial forecast of Rs. 20 crores.

Year	1	2	3	4	5	6
Actual Demand	22	25	24	28	31	33

6. Solve the following of sales forecasting using method of least square and forecast for 03.

Year	97	98	99	00	01	02
Sales	50	65	82	85	93	97

7. Following are the past data of Indian Rare Earth Ltd. Mumbai, is largely export oriented chemical industries. Exports during 1999-2003exhibited in the following table.

Year	Exports (Rs. Crores)
1999	1.19
2000	1.05
2001	1.49
2002	1.69
2003	2.90

Calculate a) exports during 2004 b) If an analysis of business conditions and economic factor indicate that exports in 2006 will be about 15% below trend or normal, and then what will be export forecast in 2006?

Reference Books:

1. PPC and Industrial Management by K C Jain and L N Agrawal
2. Industrial Engineering and Management by O P Khanna

Marks obtained:	Signature of faculty:	Date:

Production & Operation Management (ME-403.01) **Date:**

Experiment No. 3
EXERCISE ON OPERATION AND ROUTE-SHEETS

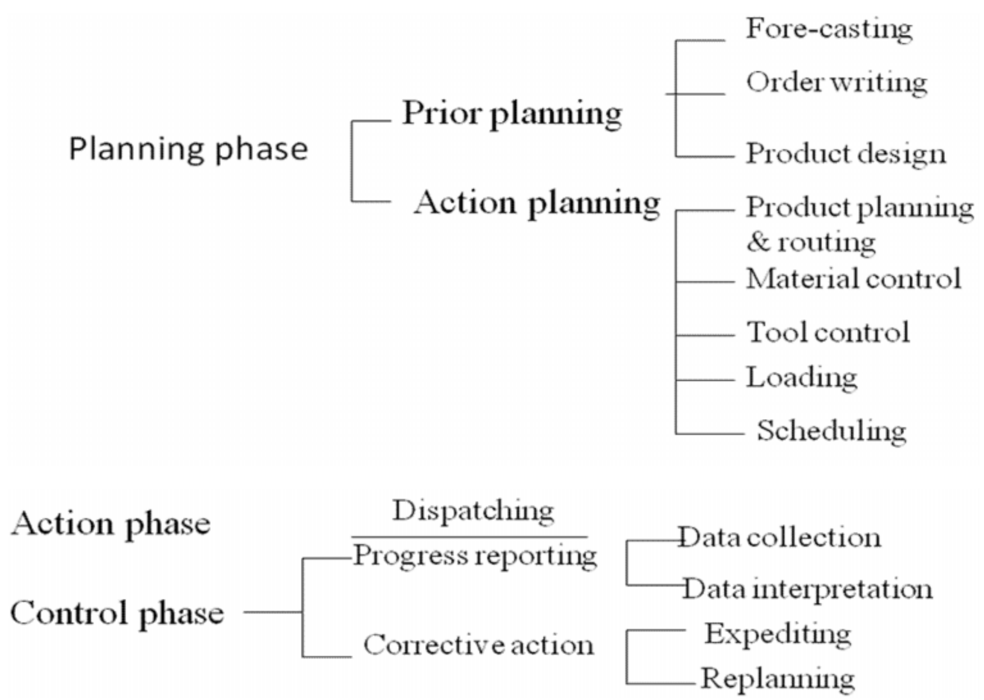
Aim: To prepare operation and route sheets based on the job given.

Objective:
- To study different functions of Production Planning and Control

Theory:

Definition of PPC
According to Samuel Eilon:
"The highest efficiency in production is obtained by manufacturing the required quantity of the product, of the required quality, at the required time, by the best and cheapest method."

Different functions of PPC



Review Questions:

1. Draw operation and route sheets for the following figures in the following format

PART NO:

NAME:

MATERIAL:

QUANTITY:

SHEET NO:

Sr No.	Sequence of Operation	Machine	Department	Tools	Jigs/Fixtures/Gauges	Time	
						Set Up Time	Operation Time

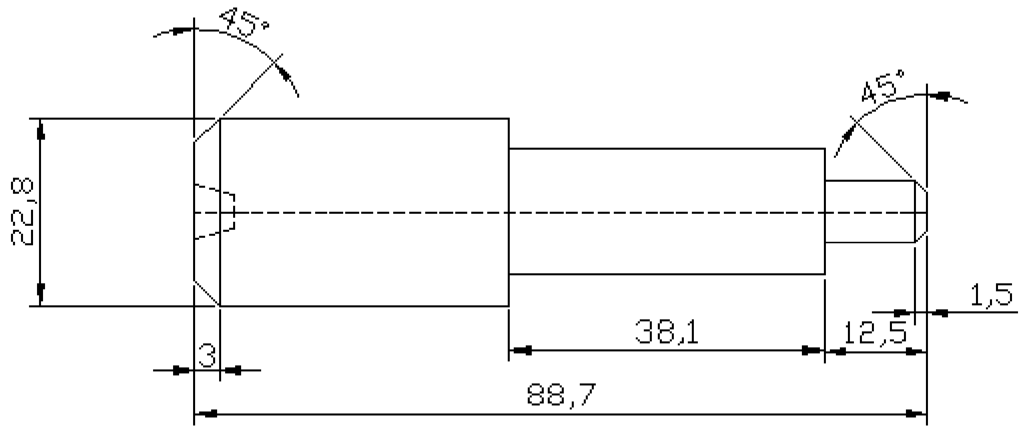


FIGURE: 1

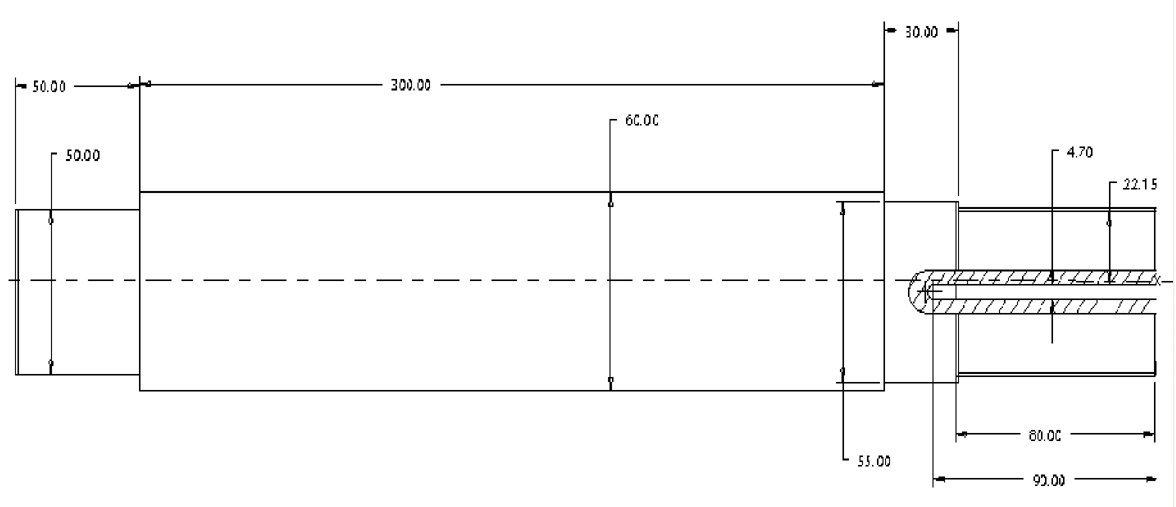


FIGURE: 2

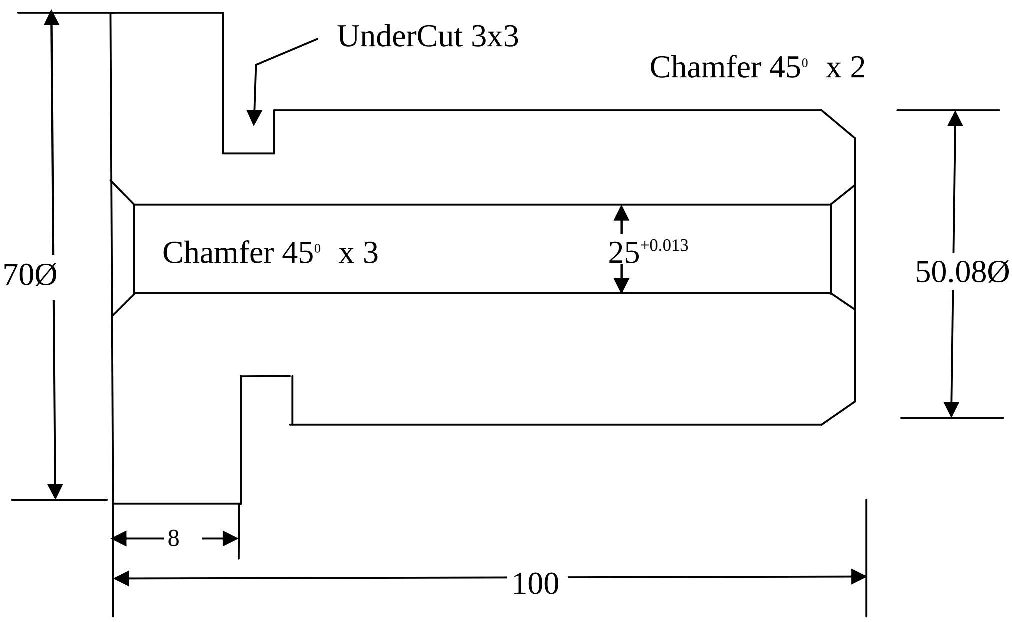


FIGURE: 3

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Experiment No. 4
PROBLEMS ON SEQUENCING TECHNIQUE, GANTT CHART

Aim: To study the Sequencing Technique and Gantt charts.

Objective:

- To study the need of sequencing.
- To learn Johnson's rule for sequencing.
- To learn to use gantt chart

Review Questions:

1. Seven jobs required to be processed through two machines A and B. The processing time (hrs) of each jobs on the two machines is given below:

Jobs	Processing Time (hrs)	
	Machine A	Machine B
1	10	5
2	20	21
3	5	4
4	25	15
5	15	14
6	12	12
7	6	9

2. Using graphical method, find the minimum elapsed time to perform jobs j1 and J2 on machines A through D, the sequence and process timings for which are given as follows:

Job 1	Sequence	**A**	B	C	D
	Time (hrs)	2	4	5	1
Job2	Sequence	B	C	D	A
	Time (hrs)	6	5	2	3

3. Solve the following sequencing problem giving an optimal solution when passing time is not allowed.

Jobs					
Machines (hrs)	A	B	C	D	E
M1	10	12	8	15	16
M2	3	2	4	1	5

M3	5	6	4	7	3
M4	14	7	12	8	10

4. Solve the following sequencing problem of jobs on is machine Mj=1,2,…6 in the order M1, M2,…, M6.

Jobs	Machines (hrs)					
	M1	M2	M3	M4	M5	M6
A	18	8	7	2	10	25
B	17	6	9	6	8	19
C	11	5	8	5	7	15
D	20	4	3	4	8	12

5. Draw the Gantt Progress Chart for the schedule of installing of bus stand involving the following activities:

Activity	**Duration (days)**	**Preceding Activities**
A. Digging foundation	2	-
B. Procuring concrete	4	A
C. Curing concrete	10	B
D. Fabricating shed	6	-
E. Transporting	2	D
F. Mounting and fixing shed	4	E
G. Procuring fitting	2	-
F. fixing sign port	2	F

Calculate total project duration and show the progress of activity at the end of the 8^{th} day on the chart, if progress of activity is as scheduled.

Reference books:

1. Operation Research by Kanti Swarup, P K Gupta and Manmohan
2. Operation Research by R K Gupta
3. Operation Research by S D Sharma
4. Operation Research by S K Jain and D M Mehta

Marks obtained:	**Signature of faculty:**	**Date:**

Production & Operation Management (ME-403.01) **Date:**

Experiment No. 5
PROBLEMS ON X-R CHARTS & OC CURVE

Aim: Experimental frequency distribution & construction of X-R charts and Proof of sampling & construction of O.C. Curve.

Objectives:

- To learn implementation of X-R chart.
- To check whether the processes is in statistical control or not.
- To determine whether the process capability is compatible with specifications.
- To understand how well a particular sampling plan is effective in discriminating between good lot & bad lot.
- To get idea of probability of acceptance of a batch having a particular percent defective items.

Theory:

Control chart is "a chronological graphical comparison of actual product quality features, with limits reflecting the ability to produce, as by the past experience on product characteristics".

As per the two methods for collecting the data through inspection, the control chart are of two types:

1. Control charts of variables
2. Control charts of attributes.

The X-R charts are classified under 'control charts of variables' in which quality is described in terms of dimensions, weights or other characteristics. These charts are most widely used control charts and present essentially a simplified method of determining the limits of variation that can be expected in the averages of small samples taken from a constant-ca7use system using the range(difference between largest & smallest value) as a measure of dispersion.

Operating Characteristics (OC) Curve is a tool to specify the power or strength of given sampling plan. It is a graphical representation of fraction defective or percent defective in a lot against the probability of acceptance.

In any acceptance sampling plan, three parameters are specified. The first parameter is number of articles N in the lot from which sample is drawn. The second parameter is the number of articles n in the random sample drawn from the lot, & the third is the acceptance number C.

This acceptance number C is the maximum allowable number of defective articles in the sample. If more than C defectives are found in a sample the lot is liable to be rejected.

Since the lot size has little affect on the probability of acceptance, therefore lot size is generally ignored in deriving sampling plan.

Review questions:

1. In a capability study of a lathe used in turning a shaft to a diameter of 23.75 ± 0.1 mm a sample of 6 consecutive pieces was taken each day for 8 days. The diameters of these shafts are given below:

1stDay	2ndDay	3rdDay	4thDay	5thDay	6thDay	7thDay	8thDay
23.77	23.80	23.77	23.79	23.75	23.78	23.76	23.76
23.80	23.78	23.78	23.76	23.78	23.76	23.78	23.79
23.78	23.76	23.77	23.79	23.78	23.73	23.75	23.77
23.73	23.70	23.77	23.74	23.77	23.76	23.76	23.72
23.76	23.81	23.80	23.82	23.76	23.74	23.81	23.78
23.75	23.77	23.74	23.76	23.79	23.78	23.80	23.78

Assuming A_2=0.48, D_3=0, D_4=2 & d_2=2.534 construct the X-R chart and find out the process capability for the machine.

2. A control chart for defects per unit u uses probability limits corresponding to probabilities of 0.975 then σ = +1.96 and 0.025 then σ = -1.96. The central line on the control chart is at u' = 2.0. The limits vary with the value of n. Determine the correct position of these upper and lower control limits when n = 5.

3. In a manufacturing process the number of defectives found in the inspection of 20 lots of 100 samples is given below:

Lot. No.	No. of defectives	Lot. No.	No. of defectives
1	5	11	7
2	4	12	6
3	3	13	3
4	5	14	5
5	4	15	4
6	6	16	2
7	9	17	8
8	15	18	7
9	11	19	6
10	6	20	4

(a) Determine the control limits of p chart and state whether the process is in control.

(b) Determine the new value of mean fraction defective if some points are out of control. Compute the corresponding control limits and state whether the process is still in control or not.

(c) Determine the sample size when a quality limit not worse than 9% is desirable and a 10% bad product will not be permitted more than three times in thousand.

4. Draw an ideal O.C curve where it is desired to accept all lots having 3% or less defectives i.e. having probability of acceptance 100%, While all lots with more than 3% defectives have a probability of acceptance 0%.
5. Draw an O.C. curve for sampling plan of n = 300, C = 10, AQL = 2%, LTPD = 5%, manufacturer's risk = 5% & Consumer's risk = 10%.
6. A single sampling plan uses a sample size of 15, and an acceptance number. Using hyper geometric probabilities, compute the probability of acceptance of lots of 50 articles 2% defective.

References:

1. Statistical Quality Control by M Mahajan
2. Quality Control by S.A.H.Khan, Z.A.Khan, D.K.Singh, G Alam
3. Industrial Organization & Engineering Economics by S.C.Sharma & T.R.Banga
4. Quality Control by Technical Teachers Training Institute-Madras
5. Production Management by L.N.Agarwal & K.C.Jain
6. Production And operation Management by L.C.Zhamb

Marks obtained:	**Signature of faculty:**	**Date:**

Experiment No. 6
VERIFICATION OF PROOF OF SAMPLING

Aim: Experiment to construct C chart and Proof of sampling

Apparatus: Plastic box and 1000 beads,
Beads distribution as follows

White :	80 %	Yellow	5.0 %
Red :	2.5%	Green	10.0%
Blue :	10%	Brown	0.5%

Theory:
Quality Control and Statistical Quality Control
The term quality control refers to the process of checking the quality of items produced with a present standard of quaiity.71t also points out defective items and processes and suggests way of reciting them. (Statistical quality control on the other hand uses statistical methods based upon mathematical theory of probability to control quality with the object of establishing quality standards and maintaining them in most economical manner. -
The SQC concept has provided a basis for determining a good or acceptable process behavior, and hence all deviations from this behavior could be traced, identified and eliminated from a process. so that the process continues to produce items of acceptable quality. The loots and techniques of sqc help in building an information system capable of providing deeper understanding of processes with the result that quality improvements could be achieved on a continuous basis.

Sampling -
Inference about sipliations have to he made continually. Samples drawn from a population are often the means used to obtain information about the populations it self. For this information to he reliable, it is necessary that the samples chose n arc representative As an example of false inference obtained on the basis of a non representative sample, we might mention an estimate of the average income based on a random sample drawn from a telephone directory

The theory of sampling provides guides to the. selection of a sample that will be representative as well as an estimate of the degree of accuracy of the inferences drawn from a representative sample of a given size.
The following are the more important ways of drawing a representative sample.

Judgments Sampling
‘this is the least expensive and most inaccurate, where a person would estimate the characteristics of a population merely by sight. Because of its inaccuracy, however, this is not frequently used.

Simple Random Sampling :
In tins type we draw a sample of a certain size from a population in such a ways as to ensure member of the population has an equal chance of being included in the sample. A convents way of using such a sample is to give a serial number to each member of the population and then make withdrawals according to a Random Number Table.

Systematic Sampling
Systematic Sampling of population is made by drawing an item at regular intervals, e.g. every fourth fifth. This can be done only when it is certain that such a sample will not be biased.

Stratified Random Sampling
Here, the population is split into groups according to same characteristic and a simple random sample drawn from each group. In this way the characteristics of the groups are obtained and hence, the characteristics of the population.

Proportional Stratified Random Sampling
This is extension of the previous method where the samples drawn from each group (n_1, n_2, n_1,) are so chose in relation to the total numbers in the groups (N_1,N_1,N), that

$$\mathbf{(n_1/N_1) = (n_2/N_2)\ \text{-------}\ (n_p/N_p) = n/N}$$

The ratio of each sample size to the group from which it is taken is the same as the ratio of the number in the total sample to the whole population.
Hither to we have only considered the distribution of a total population. However, if we take several from a total population. Having a normal distribution, each sample will have its own mean and standard deviation. W we take sufficient samples. The means of the samples will themselves form a normal distribution. The mean of this distribution will he very close to the actual true mean of the Whole population, and the more samples there are, the never will be computed mean to the true mean.

Sampling Plans
Sampling plans may be used for the acceptance (or rejection) of products or items already produced on the basis of sampling inspection. The uses of acceptance sampling are

1. To determine the acceptability of the incoming products and outgoing products.
2. 2. To determine the acceptability of the products and from one department to another within plant.
3. To select the vendors who supply specified quality materials required tar processing.
4. To improve the quality of the material supplied by the vendors.

The sampling plans may he grouped as :
1. Single sampling plan.
2. Double sampling plan.
3. Multiple sainpling plan

Procedure
1. Take the wooden, paddle and draw the sample of 50 beads.
2. Count the no. of beads of each color in the sample and write on observation table.

3. Replace the sample into box and shake thoroughly and repeat above procedure.
4. Continue this for sufficient number of samples.
5. Make the total for each color bead in the sample and find percentage assurance of cacti type of bead.
6. Find the difference between actual percentage of bead in the lot and percentage occurrence of each column
7. Comment on the result obtained.

Observation Table

Sr. No.	White	Blue	Green	Yellow	Brown	Red
1.						
2.						
3.						
4.						
5.						
6.						
7.						
8.						
9.						
10.						
11.						
12.						
13.						
14.						
15.						
16.						
17.						
18.						
19.						
20.						
21.						
22.						
23.						
24.						
25.						
26.						
27.						
28.						
29.						
30.						
Observation no. of beads						

Quality of Lot :
White :
Blue :
Green :

Yellow :
Brown :
Red :
Total :
Result Table :

Each colour

Color	Actual No of each colour Beads	% of each colour bead (given)	Observed No. of beads	% of occurrence of each colour beads	Error
White	800	80 %			
Blue	20	02 %			
Yellow	50	05%			
Brown	5	0.5%			
Red	25	02.5%			
Green	100	10%			
Total	1000	100 %			

Observed No. of beads = sample size × No. of samples drawn
= 50 × 30
Total no. of beads inspected = 1500

Sample Calculation

Observed % of white bead = Σ [(White bead x 100) / observed no, of bead]

=Summation of a particular colour beads in 30 sample/ Total No. of beads inspected in 30 beads

Error = % of defective bead (actual)
% of occurrence of each colour beads in sample

Conclusion

Marks obtained:	Signature of faculty:	Date:

Experiment No. 7
OPERATION CHARACTERISTIC CURVE (0.C.CURIVE)

Aim : Construction of operation characteristic curve (O.C curve)
Apparatus: Wooden box and 1000 beads distribution is as follows

White	80%	800
Green	l0%	100
Yellow	5%	50
Red	2.5%	25

Theory:

Every Acceptance sampling plan has an operating characteristic curve A. Following points given below emphasis the need of the operation characteristics curve.

1. There is some chance that good reality lots of materials may he rejected.
2. There is some chance that the bad till he accepted.
3. Their risk can he evaluated by the theory of probability and depends on the number of sample inspected. the acceptance number, The present delivered lots offered for sample inspection . Given the amount of risk which can be tolerated ,a sampling plan can he derived to meet the requirement.
4. The larger the sample used for inspection the nearer the operation characteristics approached the ideal. However beyond a certain point its adds the cost in inspection of a larger number of parts.

Parameters for, operation characteristics curve
The two parameters of the operation characteristics curve are the sample size and acceptance number. The desired quantity level (%) and probability of acceptance (Pa) must be selected so that the proper sampling plan can be desired.
1. ACCEPTABILITY QUALITY LEVEL (1.Q.L.):- To measure the produce’s risk. a minimum percent of defective items1in a lot is decided below which the lot should he acceptable . This is known as acceptable quality level.
2. REJECTABLE QUALITY LEVEL (R.Q.L) :-. To measure the customer risk, we must then decide the maximum percentage of defective items in a lot which can be accepted. This is known as rejectable quality level or Vol tolerance percentage defective (LTPD).
3. PRODUCERS RISK :-This is the probability of rejecting a good lot.
4. CUSTOMER’S RISK This the probability of acceptance of a bad lot.

Procedure:

1. Take the wooden padle and draw the sample of 50 beads. (Such 30 sample are to be drawn
2. Count the number of beads in sample for each colour
3. For each sampling acceptance number determine that, if the sample is accepted or noi1under three different plans i.e.c = 0 and c2.
4. Tick mark if sample is accepted h the observation table.
5. Replace the sample into the box and shake thoroughly and repeat the above procedure for nest.
6. Plot the three operating characteristics Curves from the obtained data.

Observation table (For Red colour beads) % age defective , P = 2.5 %

Sr. No	No. of table in sample e.g. Red Colour	Acceptance of sample if no. of beads		
		1	2	3
1.				
2.				
3.				
4.				
5.				
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Probability of acceptance with ZERO defective = __________________
Probability of acceptance with ONE defective = __________________

Probability of acceptance with TWO defective = ________________

Observation Table:

Sr. No	**No. of beads in sample _______Colour**	**Acceptance of sample if no. of beads**		
		0	**1**	**2**
1.				
2.				
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Probability of acceptance with ZERO defective = ________________
Probability of acceptance with ONE defective = ________________
Probability of acceptance with TWO defective = ________________

Observation Table:

Sr. No	**No. of beads in sample _______Colour**	**Acceptance of sample if no. of beads**		
		0	**1**	**2**
1.				
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Probability of acceptance with ZERO defective = __________________
Probability of acceptance with ONE defective = __________________
Probability of acceptance with TWO defective = __________________

Taking observation from proof of technique experiment and after doing exercise as above for different cases……………

Defective		**Probability of acceptance for -**		
colours	**Percentage**	**0 Defective**	**1 Defective**	**2 Defective**
Brown	0.5			
Yellow	5.0			
Blue	2.0			
Green	10.0			
Red	2.5			
Blue + Yellow	2+5=7			
Brown + Yellow	0.5+5=5			
Red + Brown	2.5+0.5 = 3.0			
Green + Yellow	10+5 = 15			
Blue + Brown	2+0.5 = 2.5			
Blue + Red	23+2.5 = 4.5			

Conclusion:

Marks obtained:	**Signature of faculty:**	**Date:**

Production & Operation Management(ME-403.01) **Date:**

Experiment No. 8
PRODUCTION AND OPERATIONS MANAGEMENT CASE-STUDIES

Aim: To perform case-analysis

Introduction to CASE1: Product Development Risks

You have the opportunity to invest INR 100 billion for your company to develop a jet engine for commercial aircrafts. Development will span 5 years. The final product costing Rs. 500 million / unit could reach a sales potential, eventually of Rs. 2500 billion. The new engine can be placed in service 5 years from now, but only if it qualifies four years from now for certification clearing commercial use and only if it meets America's Federal Aviation Administration's (FAA) ever tightening standards for noise reduction. Certification also has to be obtained from India's Director General of Civil Aviation (DGCA). There is competition from world-class manufacturers like Pratt and Whitney and Rolls Royce who are developing
Competing engines. If you decide to proceed with the project, you must also determine where the new engines will be produced and develop the manufacturing facilities. If you decline to proceed, your company could invest its resources elsewhere and based on its track record, get attractive returns.

Introduction to CASE 2: Conflict of Interests

The GM (Works) has problems with manufacturing budgets, meeting cost reduction targets, and dealing with new products manufacturing schedules. When an in-depth interview (non-directive type) was conducted between the GM (Works) and the Chairman of the Company, the GM (Works) explained that many things are happening in the Company about which he is ignorant, particularly the preparation, new product integration, etc. He agrees to the view that the Company is interested in high-growth and high-profit, but he has never been given an opportunity to review his own scheme of things and explain to the top management. The production culture of the company has never been assessed whereas the stringent rules are being directed by the finance and personnel departments. And sometimes, show cause notices are being served to supervisors and senior employees. The Company is introducing new products without assessing the capability of the manufacturing system and the resources.

Introduction to CASE 3: Service Blues!

Jyoti had given her branded laptop for servicing to an authorized service centre to repair a damaged USB port. The laptop was to be given the next day, but when she went to take it that day, she was told that it was not ready. Jyoti had to wait for four more days before she was finally given her laptop. Because she was in a hurry while receiving the repaired laptop, she did not check the workings of the laptop at that time. On reaching home and switching on the laptop, she noticed that that LCD display had become problematic. The next day, she again went to the service centre and reported the display problem. Jyoti was aghast when she was informed that as she had signed the delivery documents, the service centre cannot take responsibility for the display problem. She was asked to fill up a fresh service requisition form to get the problem rectified and further was told that all expenses incurred in rectifying the problem had to be paid by her.

Review Questions:

CASE 1:

1. What would be your line of action?
2. In case of lengthy product design and development time, what kinds of risks are there?

CASE 2:

1. Under the above situation, if you are asked to work as a consultant to show the perspectives to the Board of Management, what action plans would you suggest?
2. Does Business Process Re-engineering (BPR) help in situations like these?

CASE 3:

1. Do you think that After Sales Service through a third party is a cause for concern? Justify.
2. There seems to be a breach of trust in the given caselet. How is *breach of trust* related to quality of service?
3. In the context of the given caselet, formulate a Quality Service Policy to ensure customer satisfaction.

Reference Books:

1. Industrial Engineering and Production Management by M Mahajan
2. Industrial Engineering and Management by O P Khanna.
3. Computer aided Product Management by P B Mahapatra.
4. Production Operation Management by Panneerselvam

Marks obtained:	Signature of faculty:	Date:

Experiment No. 9
CASE-STUDY ON JIT AND INDIA

Aim: To perform case analysis on JIT

Introduction to the case:
Indian Institute of Materials Management (IIMM) is a forum for purchase and materials related employees and they have frequent meetings, seminars and annual conventions to share knowledge.

In one of their annual conventions the topic was implementation of JIT for competitive advantages. Leading personalities of the Indian industries talked lots of positive points and benefits due to the JIT purchase and JIT manufacturing methods. Many presented calculations and statistics of savings in costs and time and how it helps in reduce the price of the end products and hence competitive advantage.

Most of the audience was impressed about the theory and thought of practicalapplication in their respective companies. However, few of the executiveparticipants were mo re worried about practice and less interested in idealistictheories.One Mr. JItendra Joshi of LML's Ban galore office was impressed. He has beenarranging Engine Block castings, tyre tube sets, machined co mponents, speedometers etc from southern region to LML, Kanpur Unit. He has 15 years' of experience in facing lots of problems in arranging the long distance supplies. He mustered courage to get up and ask few questions against the JIT and summary of question to Mr. Sudhakar (the speaker) were as follows: Mr. Joshi said JIT cannot be fully implement able in Indian conditions due to following genuine constraints.

(a)The inter-state disputes like 'Kaveri Dispute', "Border disputes' at times disturb the arrangements.
(b)On and off terrorism, political agitatations, holidays due to local, regional and national leaders' deaths also disturb work environment.
(c)Spread of vendors all over India and vastness of coverage do not enable to know correct position of WIP of vendors.
(d)Transport bottlenecks, heavy rains, floods (coastal areas), workers' strikes cause anxiety and worry.
(e)Partnership problems, financial and quality constraints are not easilyattended or solved.
These questions were like a mini speech on anti JIT and the atmosphere in the auditorium got charged up Mr. Sudhakar, the speaker, gave half hearted replies to questions for which Mr. Joshi and his friends were not satisfied. Finally Mr. Sudhakar said:"The system which operate successfully in Japan may not work equally well in other countries." Only when Mr. Joshi took his seat as he felt he has made his clear on practical problems than merely going through the theory. Suddenly he seems to have won the admiration of the gathering.Prof. Rao who was chairman of the technical session gave his concluding remarks. He appreciated the ideology of JIT but advised executives to take it up step by step and ensure pragmatic views and do not over depend on JIT to fail. This he told as Indian Industrial Environment is yet to mature to take care of JIT systems in totality.

Review Questions:

1. Explain why JIT purchase works well in the developing countries

2. Do you agree with Mr. Joshi's views on constraints to JIT? Explain the correct problems in northern and eastern India.

3. Write how you feel the JIT systems can be adopted in India with an example.

Reference Books:

1. Industrial engineering and Management by O P Khanna.
2. PERT and CPM by R C Gupta
3. quantitative techniques in Management by N D Vohra
4. Operation Research by S D Sharma
5. Operation Research by Kanti Swarup, P K Gupta & Manmohan

Marks obtained:	Signature of faculty:	Date:

Production & Operation Management(ME-403.01) **Date:**

Experiment No. 10
ISO9000, 14000 CASE-STUDY

Aim: Case study based on advanced techniques ISO9000, ISO 14000.

Objective:
- To study the concepts of latest management techniques.
- To study market trends.
- How to apply particular techniques on the product and how to prepare case study report.

Review Questions:
1. Prepare a case study report.
2. Write conclusion and discussed the comments.

Reference Books:

1. Industrial Engineering and Management by O P Khanna.
2. Industrial Engineering by Martand Telsang
3. Production Operation Management S.N Chary

Marks obtained:	Signature of faculty:	Date: